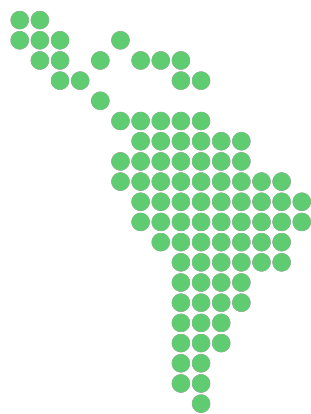


LACORO WAREHOUSE RL CHALLENGE 2025

Competition Overview and Evaluation Rubric

Simulation, Reinforcement Learning, and Social Navigation



LACORO

Latin American
Summer School
on Robotics

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Context:

Summer School LACORO 2025

Technologies:

Reinforcement Learning · ROS 2 · Gazebo Fortress

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1 Competition Context

The **LACORO Warehouse RL Challenge 2025** is a simulation-based competition focused on developing Reinforcement Learning (RL) agents capable of safely and efficiently navigating in a warehouse shared with simulated human workers (actors). The simulation is implemented in **Gazebo Fortress** and integrated with **ROS 2** to manage robot control and sensor data.

1.1 Main Objective

The main objective of the challenge is to **design and train a control policy** that enables a mobile robot to navigate between different points in the warehouse:

1. Without colliding with workers, shelves, or obstacles.
2. Respecting internal “traffic rules” inside the warehouse.
3. Completing a transport mission by visiting a sequence of stations (pickup and delivery points).

In practical terms, each team must develop an autonomous navigation policy based on RL. This policy must process the robot’s sensor data and produce safe velocity commands that allow the robot to move, avoid actors, and complete pickup–delivery missions inside the warehouse.

1.2 Motivation

The **LACORO Warehouse RL Challenge** addresses the growing need for intelligent logistics and safe human–robot interaction in Industry 4.0 environments. Participants are required to train a mobile robot using RL to:

- Safely navigate in a dynamic environment with human actors.
- Detect and avoid static and dynamic obstacles.
- Make real-time decisions that prioritize safety and mission success.

The solutions developed in this challenge are directly transferable to real-world applications such as service robotics, industrial inspection, and autonomous logistics systems, where the ability of an AI agent to move safely and reliably in complex environments is essential.

2 Target Audience and Teams

The LACORO Warehouse RL Challenge 2025 is open to students with different backgrounds and levels of expertise, from undergraduate to doctoral studies.

2.1 Target Audience

The competition is open to:

- **Undergraduate students** interested in robotics, AI, and autonomous systems.
- **Master’s students** specializing in robotics, control, machine learning, or related fields.
- **PhD students** working on advanced topics in RL, navigation, human–robot interaction, or autonomous systems.

It is recommended (but not strictly required) that participants have prior knowledge in:

- ROS 2 and Gazebo simulation.
- Reinforcement Learning fundamentals.
- Perception and sensor data processing.
- Software engineering and integration of robotic systems.

2.2 Teams

- Teams may consist of up to **4 members**.
- Interdisciplinary teams are encouraged. Ideally, each team combines skills in:
 - RL and machine learning.
 - ROS 2/Gazebo and simulation.
 - Perception and sensor processing.
 - Software engineering and system integration.

This structure encourages collaboration between students with different strengths, promotes knowledge exchange, and mirrors the kind of teamwork required in real research and industrial projects.

3 Evaluation Rubric

This section describes the rubric used by the jury to evaluate the performance and quality of each team's solution. The evaluation consists of two complementary components:

- **Model and Simulation Metrics (safety, goals reached, efficiency):** Competitors will execute their autonomous navigation model in the simulation for a duration of 5 minutes. During this time, the robot must attempt to successfully complete 5 pickup-delivery missions while avoiding collisions and maintaining efficient navigation. Once the mission finished, a brief report (e.g., `.json`, `.yaml`, `.csv`, or `.txt`) must be created summarizing key metrics, including:
 - Goals reached: e.g., 2/3.
 - Static collisions: e.g., 0.
 - Actor collisions: e.g., 1.
 - Time: e.g., 142 s.
 - Distance: e.g., 35.7 m.

Teams that generate a complete report including all metrics will receive 3 extra points. Teams that submit a report containing 3 or 4 metrics will receive 2 points. Teams that include only 1 or 2 metrics will receive 1 point. Teams that fail to submit the report will receive 0 points.

- **Model Presentation (technical explanation and innovation):** After the simulation run, each team must present the model they developed. The presentation should clearly describe:
 - The RL approach used (algorithm, architecture, etc.).
 - Reward design and shaping strategy.
 - Techniques for improving generalization and robustness.
 - Lessons learned, challenges encountered, and identified limitations.

3.1 Tolerance Objective and Proximity Thresholds

To ensure a consistent and fair evaluation of the agent's behavior, the following **tolerance thresholds** are used to define when a collision, incident, or goal completion is registered in the scoring system:

- **Distance to obstacles:** 0.5 m. Any distance smaller than this value is considered a **collision** with a static obstacle.
- **Distance to actors:** 0.7 m. Any distance smaller than this value is considered an **incident** with a human actor, and is treated as a serious safety event.
- **Distance to goal:** 0.3 m. A distance greater than this value at the end of the evaluation is considered as **not having reached the goal**.

These thresholds are used internally by the evaluation scripts and directly affect the metrics under the rubric criteria:

- *Safety and Collisions:* collisions with obstacles (distance < 0.5 m) and incidents with actors (distance < 0.7 m) contribute to penalties.
- *Task Completion:* a goal is counted as *reached* only if the robot comes within 0.3 m of the target position.

3.2 General Rubric Table

Table 1: General Evaluation Rubric

Criterion	Description	Weight	Score (0–5)
a) Safety and Actors Collisions	Automatic evaluation of interactions with human actors and static obstacles in the warehouse.	20%	
b) Safety and Obstacles Collisions	Automatic evaluation of interactions with human actors and static obstacles in the warehouse.	15%	
c) Task Completion (Navigation Objectives)	Number of goals reached, mission completion, and robustness across multiple scenarios.	15%	
d) Efficiency (Time, Path, Robustness)	Mission time, traveled distance, smoothness of motion, and consistency across different seeds.	10%	
e) Technical Design and Clarity	Quality of the RL agent design, reward function, architecture, integration with ROS 2/Gazebo, and clarity of the report/presentation.	25%	
f) Innovation and Methodological Approach	Creativity and use of advanced techniques (e.g., curriculum learning, domain randomization, hybrid RL+planning).	15%	

3.2.1 Descriptors by Criterion - Model Evaluation

The following subsections provide detailed descriptors for each criterion on a 0–5 scale.

a) Safety: Collisions with Actors (0–5 points) – Part of 20% Safety Weight

This subcriterion counts the total number of collisions with human actors across the evaluation.

Score	Descriptor
5	No collisions with actors are registered (distance to any actor is always above the 0.7 m threshold).
4	Exactly 1 collision with an actor during the evaluation set.
3	Exactly 2 collisions with actors during the evaluation set.
2	Exactly 3 collisions with actors during the evaluation set.
1	Exactly 4 collisions with actors during the evaluation set.
0	More than 4 collisions with actors during the evaluation set.

b) Safety: Collisions with Obstacles (0–5 points) – Part of 15% Safety Weight

This subcriterion counts the total number of collisions with static obstacles (shelves, walls, pallets, etc.) across the evaluation.

Score	Descriptor
5	No collisions with static obstacles are registered (distance to any obstacle is always above the 0.5 m threshold).
4	Exactly 1 collision with a static obstacle during the evaluation set.
3	Exactly 2 collisions with static obstacles during the evaluation set.
2	Exactly 3 collisions with static obstacles during the evaluation set.
1	Exactly 4 collisions with static obstacles during the evaluation set.
0	More than 4 collisions with static obstacles during the evaluation set.

c) Task Completion (Navigation Objectives, Maximum 5 points) (0–5 points) – Weight 15%

Score	Descriptor
5	Completes 5 of the navigation goals, where each goal is counted as reached only if the robot comes within 0.3 m of the target position.
4	Completes 4 of the goals under the same distance-to-goal criterion.
3	Completes 3 of the goals.
2	Completes 2 of the goals, or performance varies strongly between scenarios.
1	Completes 1 of the goals.
0	Does not complete any navigation goal in the evaluation.

d) Efficiency (Time, Path, Robustness) (0–5 points) – Weight 10%

Score	Descriptor
5	Trajectories are efficient, smooth, and stable. Mission times are low and performance is robust across all seeds.
4	Trajectories are mostly efficient, with small oscillations or slightly higher times.
3	Acceptable performance but with clearly suboptimal paths or significant variability across episodes.
2	High mission times, zig-zag motion, or erratic behavior.
1	Navigation is largely inefficient or unstable.
0	The robot frequently gets stuck or fails to make meaningful progress.

3.2.2 Descriptors by Criterion - Presentation

The following subsections provide detailed descriptors for each criterion on a 0–5 scale.

e) Technical Design and Clarity (0–5 points) – Weight 25%

Score	Descriptor
5	Very clear explanation of the RL approach, reward design (including how proximity thresholds are handled), network architecture, and ROS 2/Gazebo integration. Documentation and presentation are excellent.
4	Clear explanation, though not fully exhaustive. Documentation is good.
3	General understanding is conveyed but there are noticeable technical gaps or missing details.
2	Explanation is confusing, incomplete, or poorly structured.
1	Presentation is weak and lacks structure or clarity.
0	No technical report or presentation is delivered.

f) Innovation and Methodological Approach (0–5 points) – Weight 15%

Score	Descriptor
5	Demonstrates strong innovation through advanced techniques (e.g., curriculum learning, domain randomization, transfer learning, hybrid RL+planning) and provides deep, insightful lessons learned showing clear understanding of failures, limitations, and how design choices affected performance.
4	Presents at least one clearly innovative idea or technique, applied appropriately, and provides meaningful and well-articulated lessons learned reflecting solid understanding of the model's behavior.
3	Uses a standard RL approach implemented correctly with some minor improvements or variations. Lessons learned are present but mostly descriptive, offering limited depth or reflection.
2	Basic implementation with minimal innovation beyond typical examples. Lessons learned are shallow or incomplete, providing little analytical insight.
1	Implementation shows almost no innovation (nearly identical to baseline) and lessons learned are vague, missing, or not meaningful.
0	No identifiable methodological approach or innovation is presented, and no lessons learned are provided.